1

From the definition of k-flow from Oct 12 we have the following:

1. for each directed edge (x, y) then $f(x, y) = -f(y, x)$
2. for each vertex v, $\sum_{y \in N(v)} f(x, y) = 0$
3. for each directed edge (x, y), then $-k < f(x, y) < k$

Thus a valid 4-flow is a $\forall e \in G, f(e) = 0$. We can apply a similar principle to $Z_2 \times Z_2$ where $\forall e \in G f_2(e) = (0, 0)$.

Since 0 flows are valid 4 flows and $Z_2 \times Z_2$ flows for any graph with edges, it follows that we have a valid 4 flow iff we have a valid $Z_2 \times Z_2$ flow.

2

Proof for $\rightarrow$

Assume we have a nowhere-zero 4-flow and show we have a $Z_2 \times Z_2$ flow

Since in class we proved that we can change the direction of edges and still have a valid flow by flipping the sign of the flow. Thus we can deal with only the positive cases of $f_4(e) \in 1, 2, 3$ and we need to now care (vaguely) about the direction of edges. As a rough start, for any edge e with an odd f_4 flow, let $f_2(e) = (1, X)$ and any edge e with even f_4 flow let $f_2(e) = (0, 1)$. This is because the amount of odd flow incident edges to any vertex for the f_4 flows must be even so we can have $f_2(e)[0]$ to 1 on only the odd edges to ensure that the sum for a on a vertex to be even.

For $e \in O, O$ being the set of odd edges in f_4, the $f_2(e) = (1, X)$ has an X because there are cases where we would only have an odd amount of nodes with even flow in f_4. Thus we can define X by chooding a cycle that uses only edges in O and uses an odd one out even edge. Here an odd one out even edge would be any even edge that does not have a cycle of only even edges. So we say for $f_2(e) = (1, X)$ then $X = 1$ iff e was used in one of these cycles.

Thus we have a $Z_2 \times Z_2$ flow because odd flows must from cycles and the even flows + these special odd edges must form cycles.

Proof for $\leftarrow$

Assume we have a nowhere-zero $Z_2 \times Z_2$-flow and show we have a 4-flow

From problem 3 we have these sub graphs of even degree vertices. Thus for each subgraph we create cycles to produce a nowhere-zero 2 flow by walking in cycles around the graph and if we traverse against a directed edge then we can assign that a flow of -1 otherwise leave as $+1$. Since every vertex is of even degree we know we can create cycles that span the edges of our graph. Thus we have a nowhere-zero 2 flow for both subgraphs. I will label these flows as g_1, g_2 for the flows on G_1, G_2 respectively. We then can define $f_4(e) = g_1(e) + 2g_2(e)$ and for any edge not in $G_1, g_1(e) = 0$ similar idea for g_2.

We know that the resulting f_4 flow would be nowhere-zero because g_1, g_2 are nowhere-zero and have no way to sum together to be equal to 0 thus $f_4 \in \{\pm 1, \pm 2, \pm 3\}$. As for f_4 being a valid flow, in advanced algorithms we proved this idea of the sums of flows being a flow but I'll do a quick reproof here. If g_1, g_2 are valid flows then the sum of the flow at an individual vertex is 0 and across any cut is also 0 (by in class). Thus by combining these two flows that sum to 0 at each vertex, satisfy $f(x, y) = -f(y, x)$ and across any cut is 0 then we have a flow that satisfies all the definitions of flow.

Therefore f_4 is a valid nowhere-zero flow

3

Proof for $\leftarrow$

Suppose we have $G = G_1 \cup G_2$ where all vertices in G_1, G_2 are of even degree. Thus we define our $Z_2 \times Z_2$ flow on G as $f(e) = (I_1, I_2)$ where $I_1 = 1$ iff $e \in G_1$ else 0 and $I_2 = 1$ iff $e \in G_2$ else 0.

What this means is for any vertex $v \in G$, $\sum_{y \in N(v)} f(x, y) = (|N_1(v)| mod 2, |N_2(v)| mod 2)$ where $N_1(v)$ is neighbours of v in G_1 (and similarly for $N_2(v)$ being for G_2). Thus our value is $(0, 0)$ for a vertex iff the number of neighbours for v is even for both G_1, G_2. Since we are given that G_1, G_2 are of even degrees for every vertex, this neighbour set for all vertices would be even in size. Therefore we have created a valid $Z_2 \times Z_2$ flow for G given that G is the union of two all even degree graphs.

Proof for $\rightarrow$

I'm not convinced this actually holds for edges with flow of $(0, 0)$. A counter example being K_2 it has a valid $Z_2 \times Z_2$ flow where its edge between the two vertices is flow $(0, 0)$ but this can't be the union of two graphs of only even degrees.

Thus I am assuming that we have a nowhere-zero $Z_2 \times Z_2$ flow on G. Similar to the previous direction, we say an edge is in G_1 iff $f(e)[0] = 1$ and an edge is in G_2 iff $f(e)[1] = 1$. We know that $\forall e \in G, f(e) \neq (0,0)$ thus each edge would either be in G_1 and/or in G_2. We also know that by the definition of $Z_2 \times Z_2$ flow that the sum of flow along edges incident to a vertex should be even (so that way mod 2 of the sum is 0 for both a, b). This means that since a mod $2 = 0$ then all vertices in G_1 would have an even amount of edges by our construction of G_1 and similarly since b mod $2 = 0$ all vertices in G_2 are of even degree. We know that since we have a nowhere-zero flow that each edge would have some nonzero flow thus each edge in G would either be in G_1 and/or G_2. Thus $G_1 \cup G_2 = G$ and G_1, G_2 are all even degree vertices.